

Module 3: Logic Knowledge Representation & Resolution

Propositional Logic, FOL, Unification, Skolemization & Refutation

Module III: Knowledge Representation & Logic — Topics Covered

1. Propositional Logic Syntax & Truth Tables
2. Inference Rules & Forward/Backward Chaining
3. Propositional Resolution Refutation
4. First-Order Logic (FOL) & Quantifiers
5. Unification (MGU) & Skolemization

1. First-Order Logic Resolution Refutation

To prove $\mathbf{KB} \models \alpha$, we add $\neg\alpha$ to $\mathbf{KB}$ and derive the empty clause ($\square$) using 7 systematic CNF transformation steps:

1. Eliminate $\iff$ and $\implies$.
2. Push $\neg$ inward ($\neg\forall x P(x) \equiv \exists x \neg P(x)$).
3. Standardize variables (distinct names per quantifier).
4. **Skolemization**: Replace existential quantifiers ($\exists$) with Skolem constants or functions of outer universal variables.
5. Drop universal quantifiers ($\forall$).
6. Convert to CNF using $\vee$ distribution over $\wedge$.
7. Resolve complementary literals using **Most General Unifier (MGU)**.